
PuppetGait: Generalizing Gait Recognition via 3D Body-Aligned LVM Features

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Abstract

1 Large vision model (LVM)-based gait recognition has achieved remarkable per-
2 formance. However, current LVM-based methods largely rely on 2D image-plane
3 representations and often neglect the underlying 3D structure of the human body.
4 Without 3D structural guidance, 2D representations are sensitive to viewpoint varia-
5 tions. This sensitivity leads to unreliable gait features and limits generalizability in
6 real-world 3D scenarios. To bridge this gap, we propose **PuppetGait**, a framework
7 that aligns LVM features into a unified 3D canonical body space. The method
8 conceptually manipulates the features like a puppet, adjusting them to a prede-
9 fined pose and viewpoint. Comprehensive evaluations on CCPG, CCGR-MINI,
10 CASIA-B*, and SUSTech1K show that PuppetGait maintains strong within-domain
11 performance (96.54% mean Rank-1 on CCPG) while substantially improving cross-
12 domain generalization, reaching 23.22% Rank-1 on CCPG-to-CCGR-MINI and
13 63.28% mean Rank-1 on CCGR-MINI-to-CCPG. Code will be publicly available.

14 1 Introduction

15 Gait recognition identifies individuals by their unique walking patterns, offering a non-intrusive
16 biometric modality for long-distance and non-cooperative identification.[13, 26, 33]. These properties
17 make gait recognition valuable for security applications such as surveillance-based retrieval, suspect
18 tracking, and identity verification, where subjects may be observed by ordinary cameras without active
19 cooperation. Despite substantial progress in RGB-based gait analysis, robust cross-view recognition
20 remains a fundamental challenge. This critical issue has been explicitly studied since early multi-view
21 benchmarks such as CASIA-B [50] and OU-MVLP [39], and becomes even more entangled with
22 clothing, carrying conditions, and camera-domain shifts in recent practical benchmarks like CCGR
23 [55].

24 Prior works have explored cross-view challenges through two main routes as illustrated in Figure 1.
25 Most existing methods learn view-invariant descriptors from 2D observations [25, 2, 6, 21]. These
26 methods are practical and compatible with camera inputs, but their representations remain tied to
27 the observed image plane, so cross-view body-part correspondence is mostly learned implicitly from
28 training data. In contrast, structure-aware methods introduce pose [20], parametric body meshes
29 [51], LiDAR point clouds [34], or explicit 3D galleries [56]. While these methods provide a stronger
30 basis for cross-view correspondence, they often impose additional constraints, such as sparse pose
31 representations, template-confined meshes, dedicated depth sensors, or subject-specific 3D enrollment.
32 This reveals a key trade-off: image-plane methods preserve practical RGB pipelines but lack explicit
33 spatial alignment, whereas explicit 3D methods improve structural correspondence but are less
34 flexible as general RGB recognition solutions.

35 Recent advances in pretrained vision foundation models provide powerful tools for gait recognition.
36 On the representation side, LVM-based gait methods have introduced strong pretrained visual features

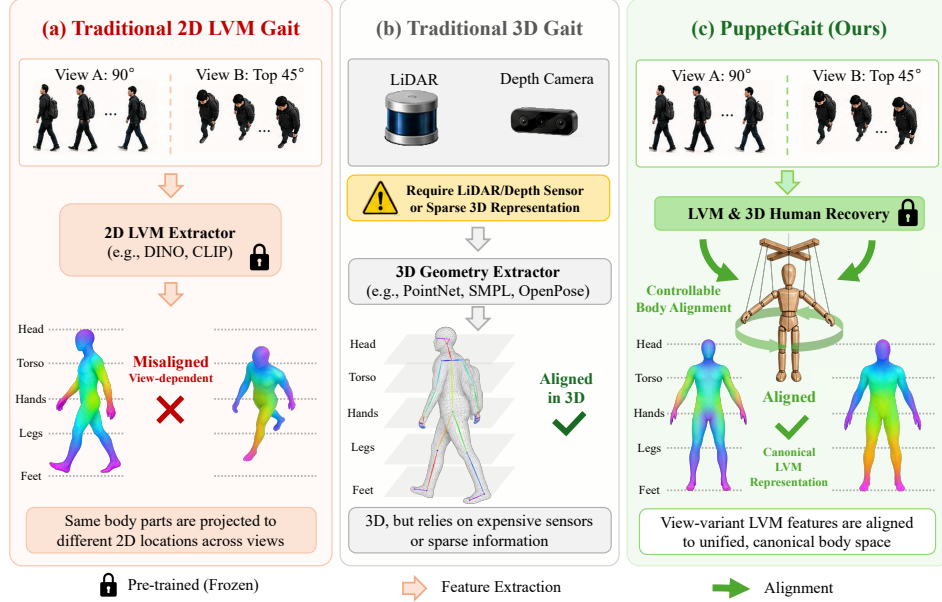


Figure 1: Overview and motivation of PuppetGait. Traditional 2D LVM methods preserve rich RGB semantics but remain view-dependent in the image plane. Traditional 3D methods are inherently aligned but rely on depth/LiDAR sensors or sparse 3D representations. PuppetGait uses LVM inferred 3D body to reorganize LVM features into a canonical feature layout.

into gait modeling. BigGait[47] adapts task-agnostic LVM features for gait recognition, while BiggerGait[48] further shows that intermediate LVM layers contain discriminative and complementary identity cues. However, existing LVM-based methods primarily exploit feature discriminability within the standard 2D recognition pipeline, without introducing an explicit alignment mechanism. As a result, their robustness to viewpoint changes and domain shifts still largely depends on learned invariance from large-scale data, rather than explicit body-part correspondence. Meanwhile, LVM-based human 3D mesh recovery model[46] can infer dense, camera-aware body geometry from ordinary RGB frames, including body shape, articulated pose, and camera configuration. This geometry is represented by a parametric human body model, enabling the recovered body to be controlled by changing its pose and rendering viewpoint while preserving the inferred body shape.

This motivates us to revisit a long-standing intuition in gait recognition: reducing spatial variation by mapping human body toward a common view or canonical layout. Unlike view-transformation methods that operate on 2D gait templates, we apply this alignment to **dense LVM feature maps with the help of 3D body geometry**. Specifically, the parametric body recovered by a 3D human mesh recovery model serves as a controllable puppet: we first anchor image-plane LVM features to the inferred body and then manipulate the body into a preset target configuration, such as an A-pose rendered from a fixed view.

Following this insight, we introduce **PuppetGait**, a controllable body-alignment framework for LVM-based gait recognition. PuppetGait leverages a frozen LVM-based 3D human mesh recovery model to extract LVM features and estimate body geometry. The inferred body geometry is reconfigured into a preset target configuration to establish consistent body-part correspondence across viewpoints, as illustrated in Fig. 1(c). Guided by Geometry-aware Optimal Transport (GAOT), the original view-dependent 2D features are mapped into a canonical feature layout. This design preserves rich LVM semantics while improving cross-view comparability and cross-domain generalization.

To the best of our knowledge, this is the first LVM-based RGB gait framework that uses RGB-inferred 3D human geometry as a structural prior for feature-level body alignment. Our main contributions are summarized as follows:

- **A 3D-Guided Canonical LVM Framework:** We identify the spatial misalignment inherent in view-dependent LVM features and propose to use LVM-inferred body geometry as a

structural carrier rather than an independent modality. Through Puppet Alignment and GAOT, PuppetGait anchors 2D image-plane LVM features to the inferred body surface and transports them into a canonical body-aligned feature layout.

- **State-of-the-art performance and strong generalization.** Extensive experiments on CCPG, CCGR-MINI, CASIA-B*, and SUSTech1K show that PuppetGait establishes state-of-the-art results in both within-domain and cross-domain settings. The gains are especially pronounced under cross-domain evaluation, demonstrating the effectiveness of 3D-guided canonical alignment for generalizable gait recognition.

2 Related Work

2.1 Gait representation learning

Early gait recognition relies on handcrafted body templates, including gait energy images and view transformation models [11, 25]. Modern deep methods typically learn sequence-level representations from silhouettes, with representative models improving set modeling, part-level temporal modeling, and global-local representation [2, 6, 21]. OpenGait and GaitBase further provide practical benchmarks and strong baselines for evaluating architectural progress [7, 9]. Beyond silhouettes, pose, skeleton, and parsing-based representations have been explored to inject human structure into gait recognition [20, 42, 8]. Mesh-based gait methods provide a denser body representation for structural reasoning [51, 44]. These methods demonstrate that body structure matters, but they usually commit to a fixed upstream representation. PuppetGait instead keeps rich LVM features and uses inferred body geometry only to establish controllable body-part correspondence, rather than another input modality.

2.2 Large vision models and human mesh recovery

Foundation models have changed visual representation learning by pretraining transferable backbones on large-scale data. CLIP learns visual features from language supervision [31], DINO-style models learn self-supervised visual features with strong transfer [1, 27, 36], and SAM provides promptable segmentation priors [17]. BigGait introduces LVM features to gait recognition by transforming all-purpose features into implicit gait representations, while BiggerGait shows that intermediate LVM layers contain complementary gait information [47, 48]. Other recent works explore denoising, large-language-model sequence modeling, and multi-modal gait feature construction [15, 45, 14]. In parallel, human mesh recovery and 3D body models have advanced from SMPL/SMPL-X based parametric bodies [23, 28] to transformer and promptable full-body recovery models [35, 5, 46]. PuppetGait connects these two trends by using downstream inferred body geometry to control the alignment of upstream dense LVM gait features.

2.3 Cross-view, cross-covariate, and multi-modal gait

Practical gait recognition must handle cross-view and cross-covariate variations, including clothing, carrying, occlusion, speed, and dataset domain shifts [50, 39, 18, 55]. In-the-wild benchmarks such as GREW and Gait3D reveal further domain complexity, while CCGR emphasizes mixed covariates beyond isolated view or clothing changes [54, 51, 55]. Recent methods address these issues through dynamic temporal representations, quality-aware modeling, causal intervention, and cross-granularity alignment [41, 43, 52, 40]. Multi-modal and 3D gait methods provide another path by using LiDAR, pointclouds, SMPL meshes, or 3D galleries [34, 14, 56]. These works suggest that geometry can reduce view ambiguity, but many require additional sensors, fixed upstream modalities, or explicit 3D enrollment. PuppetGait targets the same geometric need with standard RGB walking videos and an off-the-shelf 3D human prior.

2.4 Long-term human recognition and feature transport

Gait recognition is related to person re-identification (ReID), especially under clothing changes and long-term retrieval. ReID benchmarks and baselines have studied scalable retrieval, part pooling, transformer features, video aggregation, and re-ranking [53, 24, 12, 49]. Recent clothes-changing and body-shape ReID works further emphasize that identity cues should be disentangled from appearance

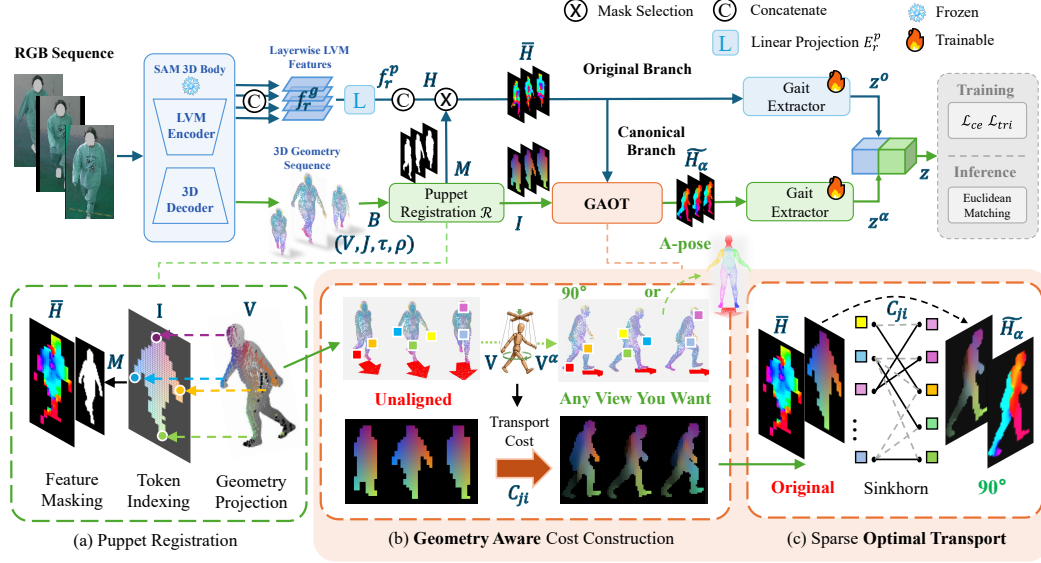


Figure 2: Overview of PuppetGait. A frozen SAM 3D Body model provides layer-wise LVM features and inferred 3D body geometry. The original-view branch encodes masked image-plane features, while the canonical branch anchors tokens to the visible body surface and transports them to a canonical feature layout by Geometry-aware Optimal Transport (GAOT). The original-view and canonical descriptors are fused for recognition.

and clothing, often through gait, pose, shape, or semantic priors [10, 16, 18, 22]. Unlike ReID methods that primarily retrieve single images or short video tracks from appearance, PuppetGait focuses on sequence-level gait features and their body-part correspondence across views. GAOT is inspired by optimal transport, which provides a principled way to match distributions or features under structural constraints [37, 4, 3, 30]. We use it for geometry-aware body-part feature transport rather than for generic domain adaptation.

3 Method

Given an RGB gait sequence, PuppetGait learns a sequence-level identity descriptor for retrieval using identity supervision during training. As illustrated in Fig. 2, PuppetGait converts LVM features into a canonical feature representation with a two-branch pipeline. For each RGB frame, a frozen SAM 3D Body model provides layer-wise LVM features and inferred body geometry. The original-view branch encodes masked image features, while the canonical branch treats the inferred body as a controllable puppet: it anchors valid LVM tokens to the visible body surface by **Puppet Alignment**. Then the **Geometry-aware Optimal Transport (GAOT)** module reconfigures the body under a preset target configuration and transports the anchored features to a canonical feature layout. Finally, both descriptors are fused for final recognition.

3.1 LVM Feature and Geometry Extraction

For each frame $\mathbf{x}$, the frozen SAM 3D Body image encoder provides selected intermediate LVM token features $\{\mathbf{F}^l\}$ and the last layer as encoder representation $\mathbf{e}$ used by its 3D decoder. Following BiggerGait [48], we group these layer-wise features by encoder depth and project each group to a compact feature map. This extraction and grouping stage corresponds to the left part of Fig. 2, where the frozen encoder supplies layer-wise LVM features and the decoder supplies the body geometry used by the canonical branch. For the r -th group, we denote the concatenated group feature and its projected feature as

$$\mathbf{f}_r^g = \text{Concat}_{l \in \mathcal{G}_r} \mathbf{F}^l, \quad \mathbf{f}_r^p = E_r^p(\mathbf{f}_r^g), \quad (1)$$

where $\mathcal{G}_r$ is the layer set of group r and E_r^p is a lightweight projection head. The projected group features are concatenated to form the compact LVM feature map. This feature map is shared by the

original-view and canonical branches.

$$\mathbf{H} = \text{Concat}_{r=1}^{N_g} \mathbf{f}_r^p, \quad \mathbf{H} \in \mathbb{R}^{C \times H_c \times W_c}. \quad (2)$$

From LVM feature map, the frozen SAM 3D Body decoder model also predicts the inferred 3D body geometry and camera settings required for alignment. Given the image-encoder representation of the same frame, its 3D decoder estimates a camera-aware human body:

$$\mathcal{B} = D_{3D}(\mathbf{e}) = (\mathbf{V}, \mathbf{J}, \boldsymbol{\tau}, \boldsymbol{\rho}), \quad (3)$$

where $\mathbf{V} = \{\mathbf{v}_m\}_{m=1}^M$ are vertex-indexed body surface points, $\mathbf{J}$ are 3D body keypoints, $\boldsymbol{\tau}$ is the predicted camera translation, and $\boldsymbol{\rho}$ is the estimated global body rotation. The geometry is used only as a structural guide for masking and feature alignment, not as an independent identity representation.

3.2 Puppet Alignment

To reorganize image-plane LVM features into a canonical feature layout via 3D body geometry, we need to associate 2D feature locations with body-surface points on the inferred mesh. Puppet Alignment is designed to establish this feature-to-body association. Meanwhile, the same projection process naturally provides a body mask, which is used to suppress identity-irrelevant background responses and retain human foreground features.

As illustrated in Fig. 2(a), we use a forward projection strategy. Given the decoded body vertices $\mathbf{V}$, camera translation $\boldsymbol{\tau}$, and camera intrinsics $\mathbf{A}$, we render the vertices onto the recognition feature grid:

$$\mathbf{I}, \mathbf{M} = \mathcal{R}(\mathbf{V}, \boldsymbol{\tau}, \mathbf{A}), \quad (4)$$

where $\mathcal{R}(\cdot)$ denotes z-buffered vertex rendering. When multiple vertices are projected to the same grid cell, we keep the vertex with the smallest camera-space depth z , corresponding to the visible surface point closest to the camera along that viewing ray. The resulting index map $\mathbf{I}$ records the visible vertex projected to each grid cell: $\mathbf{I}(i) = m$ means that feature location i is associated with body vertex $\mathbf{v}_m$. The body mask $\mathbf{M}$ indicates whether a grid cell is covered by the projected body, i.e., $\mathbf{M}(i) = 1$ if at least one body vertex is rendered to cell i , and $\mathbf{M}(i) = 0$ otherwise.

We apply the mask to the LVM feature map: $\bar{\mathbf{H}} = \mathbf{H} \odot \mathbf{M}$, where $\bar{\mathbf{H}}$ denotes the foreground-preserved LVM feature map, while $\mathbf{I}$ provides the feature-to-body anchoring used for subsequent canonical alignment.

3.3 Geometry-aware Optimal Transport

Geometry-aware Optimal Transport (GAOT) canonicalizes the source LVM features in two steps, as shown in Fig. 2(b,c): geometry-guided cost construction and sparse optimal transport. The former builds source-to-target transport costs from the canonical puppet projection, while the latter softly transports LVM features into the canonical feature layout.

Geometry-guided cost construction. We first separate grid indices, 2D grid coordinates, and 3D vertex indices. Let Ω^s and Ω^c denote the source and canonical feature grids. A source token is indexed by $i \in \Omega^s$ and has a source-grid coordinate $\mathbf{u}_i^s \in [-1, 1]^2$; a canonical target cell is indexed by $j \in \Omega^c$ and has coordinate $\mathbf{u}_j^c \in [-1, 1]^2$. Mesh vertices use a different index $m \in \{1, \dots, M\}$ and have 3D coordinates $\mathbf{v}_m$. Puppet Alignment provides a token-to-vertex map $\mathbf{I}$, where $\mathbf{I}(i) = m$ means that source token i is attached to visible vertex m after z-buffered source-view rendering.

The recovered mesh and the puppet are generated from a shared parametric template, so their vertex ordering is preserved. Thus, a vertex index can be used as a stable body-surface pointer rather than as an unordered point identifier.

For a preset target configuration α , the canonical puppet is obtained as

$$\mathbf{V}^\alpha = \mathcal{P}(\mathcal{S}(\mathbf{V}); \boldsymbol{\omega}_\alpha, \boldsymbol{\pi}_\alpha), \quad (5)$$

where $\mathcal{S}(\mathbf{V})$ denotes the body-shape component extracted from the inferred mesh, $\boldsymbol{\omega}_\alpha$ specifies the target rendering direction such as yaw and pitch, and $\mathcal{P}(\cdot)$ reconstructs the mesh under this puppet configuration. Optionally, $\boldsymbol{\pi}_\alpha$ specifies a preset body pose. The output remains an indexed vertex set,

184 $\mathbf{V}^\alpha = \{\mathbf{v}_m^\alpha\}_{m=1}^M$. Therefore, if a source token is anchored to vertex index m , the indexed lookup $\mathbf{v}_m^\alpha$
 185 gives the 3D position of that body point on the canonical puppet. In Fig. 2(b), we visualize a 90°
 186 horizontal target view without changing the walking pose.

187 For a valid source token i , the source-to-canonical lookup is

$$m_i = \mathbf{I}(i), \quad \mathbf{v}_{m_i}^\alpha = \mathbf{V}^\alpha[m_i], \quad \mathbf{p}_i^\alpha = \Pi^c(\mathbf{v}_{m_i}^\alpha), \quad (6)$$

188 where Π^c projects a 3D point on the canonical puppet to a normalized 2D coordinate on the canonical
 189 feature grid. Thus $\mathbf{p}_i^\alpha \in [-1, 1]^2$ is the canonical-grid location where the body point observed by
 190 source token i is expected to appear. Rendering the canonical puppet with the same projection also
 191 yields the canonical valid mask $\mathbf{M}^c$, where $\mathbf{M}_j^c = 1$ if target cell j is covered by the projected puppet.

192 The geometry-aware transport cost between target cell j and source token i is then defined as

$$\mathbf{C}_{ji} = \|\mathbf{u}_j^c - \mathbf{p}_i^\alpha\|_2^2. \quad (7)$$

193 The cost is measured on the canonical grid after routing source token i through its attached 3D vertex.
 194 A smaller value means that source token i is geometrically expected to contribute to target location j .

195 **Sparse optimal transport.** Direct source-to-target assignment is unstable on the canonical grid,
 196 since self-occlusion and discretization can map multiple source tokens to the same or nearby target
 197 cells, causing collisions and loss of LVM semantics. We therefore formulate canonicalization as
 198 sparse soft transport, where the geometry-aware cost $\mathbf{C}$ guides valid source tokens to canonical target
 199 cells under the source and target masks, as illustrated in Fig. 2(c).

200 Specifically, we build a pairwise cost matrix between target grid locations and source tokens, mask
 201 out invalid entries outside $\mathbf{M}$, $\mathbf{M}^c$, or the distance threshold, and retain only the top- k nearest valid
 202 source candidates for each target location j :

$$\mathcal{S}_j = \text{TopK}_i(-\mathbf{C}_{ji}, k), \quad k = 4. \quad (8)$$

203 This step defines the sparse support before Sinkhorn normalization. We then build a masked transport
 204 kernel on this support:

$$\mathbf{K}_{ji} = \exp(-\mathbf{C}_{ji}/\varepsilon) \cdot \mathbb{I}[i \in \mathcal{S}_j] \cdot \mathbb{I}[\mathbf{C}_{ji} < \delta^2] \cdot \mathbf{M}_i \cdot \mathbf{M}_j^c, \quad (9)$$

205 where ε is the transport temperature. Sinkhorn normalization gives the sparse transport plan:

$$\mathbf{P} = \text{Sinkhorn}(\mathbf{K}). \quad (10)$$

206 The canonicalized feature at target location j is obtained by

$$\tilde{\mathbf{h}}_j^\alpha = \sum_{i \in \mathcal{S}_j} \mathbf{P}_{ji} \bar{\mathbf{h}}_i. \quad (11)$$

207 After reshaping all target locations back to the canonical grid, we obtain the canonical feature map
 208 $\tilde{\mathbf{H}}^\alpha \in \mathbb{R}^{C \times H_c \times W_c}$. This operation transports the original LVM semantics according to geometry-
 209 aware correspondences, producing a canonical feature layout for gait encoding.

210 3.4 Sequence Encoding and Optimization

211 The original-view and canonical feature sequences are encoded by two GaitBase-style sequence
 212 encoders and fused by concatenation:

$$\mathbf{z}^o = G_\theta^o\left(\{\bar{\mathbf{H}}_t\}_{t=1}^T\right), \quad \mathbf{z}^\alpha = G_\theta^\alpha\left(\{\tilde{\mathbf{H}}_t^\alpha\}_{t=1}^T\right), \quad \mathbf{z} = \text{Concat}(\mathbf{z}^o, \mathbf{z}^\alpha). \quad (12)$$

213 The model is trained with the standard triplet and identity classification losses:

$$\mathcal{L} = \mathcal{L}_{\text{tri}} + \mathcal{L}_{\text{ce}}, \quad (13)$$

214 while the 3D foundation model remains frozen. During inference, probe and gallery descriptors are
 215 compared using Euclidean distance.

Table 1: Main comparison across source and target datasets. Yellow cells are within-domain results; others are cross-domain results. Best and second-best results are bolded and underlined. CCGR* denotes CCGR-MINI, SAM 3D Body* denotes its LVM encoder [46], and Avg is computed over available entries.

Train Dataset	Input	Method	LVM	Venue	CCPG [18]				CCGR* [55]		SUSTech1K [32]		CASIA-B* [50]				Avg
					CL	UP	DN	BG	R-1	R-5	CL	R-1	NM	BG	CL	Avg	
CCPG	Sil.	GaitSet [2]	–	PAMI’22	60.2	65.2	65.1	68.5	2.4	6.9	8.2	12.8	47.4	40.9	25.8	36.7	
	Sil.	GaitPart [6]	–	CVPR’20	64.3	67.8	68.6	71.7	2.4	6.9	8.1	13.5	51.2	41.9	26.0	38.4	
	Sil.	GaitGL [21]	–	ICCV’21	68.3	76.2	67.0	76.7	3.3	8.4	25.4	33.6	63.1	58.5	46.3	47.9	
	Sil.	GaitBase [7]	–	CVPR’23	71.6	75.0	76.8	78.6	2.8	7.3	9.5	16.8	59.1	52.7	30.4	43.7	
	Sil.	DeepGaitV2 [9]	–	TPAMI’25	78.6	84.8	80.7	89.2	3.7	9.1	27.0	38.4	74.6	67.2	50.2	54.9	
	Sil.+Skel.	BiFusion [29]	–	MTA’23	62.6	67.6	66.3	66.0	–	–	–	–	–	–	–	–	
	Sil.+Skel.	SkeletonGait++ [8]	–	AAAI’24	79.1	83.9	81.7	89.9	–	–	–	–	–	–	–	–	
	Sil.+Par.	XGait [52]	–	MM’24	72.8	77.0	79.1	80.5	–	–	–	–	–	–	–	–	
	Sil.+Par.+Flow	MultiGait++ [14]	–	AAAI’25	83.9	89.0	86.0	91.5	–	–	–	–	–	–	–	–	
	RGB+Sil.	GaitEdge [19]	–	ECCV’22	66.9	74.0	70.6	77.1	–	–	8.9	19.6	66.5	58.7	44.8	–	
	RGB	BigGait [47]	DINOv2-S [27]	CVPR’24	82.6	85.9	87.1	93.1	7.4	16.3	43.7	56.4	77.4	71.5	33.6	59.5	
	RGB	BiggerGait [48]	DINOv2-S	NeurIPS’25	89.8	92.1	93.7	97.5	15.5	<u>26.9</u>	70.9	79.6	93.0	90.8	55.6	73.2	
	RGB	BiggerGait [48]	DINOv2-L	NeurIPS’25	92.8	94.6	95.7	98.2	15.7	<u>26.9</u>	76.4	85.3	95.4	93.7	62.2	76.1	
	RGB	PuppetGait	SAM 3D Body*	Ours	95.5	95.7	97.0	<u>98.0</u>	23.2	37.0	<u>74.4</u>	86.8	96.1	93.8	72.6	79.1	
CCGR*	Sil.	GaitBase [7]	–	CVPR’23	20.8	31.3	<u>38.3</u>	70.2	21.1	37.8	28.7	48.0	66.3	57.7	33.9	41.3	
	Sil.	DeepGaitV2 [9]	–	TPAMI’25	23.0	37.1	36.5	69.9	26.4	45.2	32.1	51.7	72.0	62.0	36.6	44.8	
	RGB	BigGait [47]	DINOv2-S	CVPR’24	22.9	42.2	25.0	80.5	88.0	95.9	71.9	85.6	90.1	87.9	58.4	68.0	
	RGB	BiggerGait [48]	DINOv2-S	NeurIPS’25	23.2	44.5	29.5	86.7	85.7	94.1	82.1	93.8	97.2	96.4	66.7	72.7	
	RGB	BiggerGait [48]	DINOv2-L	NeurIPS’25	25.0	47.9	31.0	88.5	93.0	97.7	94.1	<u>97.5</u>	97.4	96.6	74.6	76.7	
	RGB	PuppetGait	SAM 3D Body*	Ours	44.7	72.2	38.7	97.5	94.0	97.7	<u>94.0</u>	98.1	98.4	98.1	83.9	83.4	

4 Experiments

4.1 Datasets and Evaluation Protocols

We evaluate PuppetGait on four RGB gait benchmarks in both within-domain and direct cross-domain settings. CCPG [18] is the primary clothing-changing benchmark; we report Rank-1 on full clothing change (CL), upper-clothing change (UP), pants change (DN), and bag carrying (BG), excluding identical-view matches. For CCGR [55], we use the official CCGR-MINI split, denoted as CCGR*, and report Rank-1/Rank-5. SUSTech1K [32] and CASIA-B* [50] are additional transfer targets; we report SUSTech1K overall Rank-1/CL Rank-1 and CASIA-B* NM/BG/CL Rank-1.

4.2 Implementation Details

Unless otherwise specified, PuppetGait uses SAM 3D Body as the frozen LVM-based 3D human foundation model. We use its image encoder for layer-wise LVM features and its decoder for inferring body geometry. The last 16 encoder layers are split into four groups and projected to a 64×32 feature map. For GAOT, the temperature is 0.01, the distance threshold is 0.2, top- $k = 4$, and sparse Sinkhorn runs for 4 iterations. the canonical target configuration is A-pose rendered at 0° yaw.

All models use triplet and identity classification losses and are optimized by SGD with learning rate 0.1, momentum 0.9, and weight decay 5×10^{-4} . Using four 48GB NVIDIA A40 GPUs, PuppetGait is trained for 20k iterations on CCPG and 30k on CCGR*, with learning-rate decay at 10k/15k and 15k/25k iterations, respectively. Each batch contains 8 identities and 4 sequences per identity; we sample 30 frames for training and use up to 250 frames for evaluation.

4.3 Main Results

Table 1 compares PuppetGait with representative silhouette-based, multimodal, and RGB-based methods. Cross-domain entries are omitted when target-domain modalities or annotations are unavailable.

Cross-domain Evaluation. The cross-domain columns of Tab. 1 highlight the transferability of PuppetGait. Trained on CCPG, PuppetGait yields consistent gains on every target dataset (CCGR*, SUSTech1K, CASIA-B*), establishing state-of-the-art cross-domain performance. The headline result is a CCGR* Rank-1 of 23.2%, exceeding the previous best by +7.5% under complex covariate

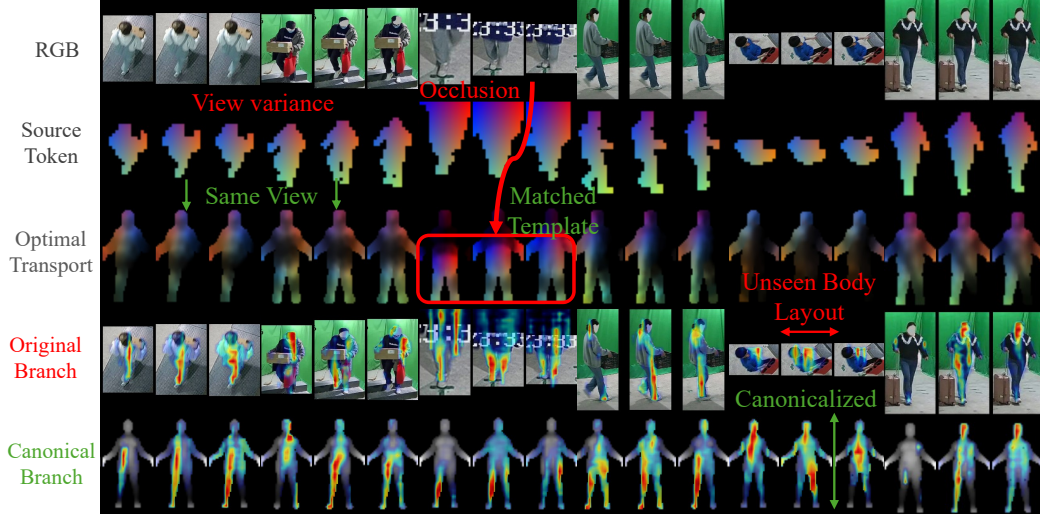


Figure 3: Visualization of PuppetGait. Rows show RGB frames, color-coded source LVM tokens, tokens after Geometry-aware Optimal Transport (GAOT), original-view branch responses, and canonical branch responses. Source tokens remain tied to the observed image layout and can become spatially inconsistent under occlusion or view changes. GAOT routes visible tokens through Puppet Alignment and transports them to a canonical feature layout.

conditions; the largest single-probe gain occurs on CASIA-B* under the clothing-change (CL) condition (+10.4%), indicating that the aligned representation effectively mitigates appearance variation. *Trained on CCGR**, PuppetGait raises the overall row average from 76.7% to 83.4% over BiggerGait-DINOv2-L and reaches 63.28% mean Rank-1 on CCGR*-to-CCPG, improving transfer to CCPG by +15.2% on average. The largest gains appear on the most appearance-sensitive CCPG probes, CL (+19.7%) and UP (+24.3%), which supports our central claim that 3D-guided canonical alignment reduces dependence on observed image layouts and yields more generalizable gait embeddings.

Within-domain recognition. Beyond transfer, PuppetGait also preserves strong source-domain performance, as shown by the yellow cells in Tab. 1. On CCPG, PuppetGait achieves 96.5% mean Rank-1, surpassing the previous best method and improving the CL by +2.7%; this indicates that canonical alignment strengthens RGB LVM features under appearance variation. Compared with MultiGait++, PuppetGait further improves CCPG mean accuracy by +8.9% using only RGB input, indicating that explicit 3D-guided alignment can provide stronger identity cues without silhouettes, parsing, skeletons, or optical flow at inference. The same conclusion holds on CCGR*, where PuppetGait obtains 94.0% Rank-1 and surpasses BigGait, BiggerGait-DINOv2-S, and BiggerGait-DINOv2-L by +6.0%, +8.3%, and +1.0%, respectively, showing strong adaptation to complex settings.

4.4 Visualization

Figure 3 visualizes the feature reorganization produced by GAOT. Source LVM tokens contain rich semantics but follow the observed camera layout, so body parts can shift or disappear under viewpoint changes, carrying, and occlusion. After transport, tokens are reorganized into a canonical feature layout, making corresponding body regions more comparable across sequences. The branch responses further show the intended complementarity: the original-view branch keeps discriminative image-plane evidence, while the canonical branch produces more stable aligned body-part activations.

4.5 Ablation Study

Table 2 summarizes four design checks under CCPG training. These ablations separate the source of LVM appearance features, the quality of inferred body geometry, the transport strategy, and the

Table 2: Ablations under CCPG training. CCPG denotes the source-domain mean Rank-1 over CL/UP/DN/BG, and CCGR* denotes direct cross-domain Rank-1 on CCGR-MINI.

(a) LVM Models and Canonical Branch				
Method	LVM source	Canonical	CCPG	CCGR*
BiggerGait	DINOv2-S	✗	93.3	15.5
PuppetGait	DINOv2-S	✓	92.3	17.9
BiggerGait	SAM 3D Body	✗	96.4	18.7
PuppetGait	SAM 3D Body	✓	96.5	23.2

(b) 3D Body Geometry					
Geometry	CL	UP	DN	BG	CCPG
ROMP [38]	93.4	93.5	96.2	97.7	95.2
SAM 3D Body	95.5	95.7	97.0	98.0	96.5

(c) Sparse Optimal Transport				
Transport	OT	TopK4	CCPG	CCGR*
Direct	✗	✗	96.4	22.9
Dense OT	✓	✗	95.9	21.9
Sparse OT	✓	✓	96.5	23.2

(d) Canonical Target Layout						
Layout	CL	UP	DN	BG	CCPG	CCGR*
w/o layout	95.7	95.6	96.3	97.9	96.4	18.7
w/o A-pose, 0°	95.4	95.9	96.8	97.9	96.5	23.0
w/o A-pose, 90°	95.5	95.6	96.3	98.1	96.4	23.1
A-pose, 0°	95.5	95.7	97.0	98.0	96.5	23.2

canonical target layout, verifying that the transfer gain comes from 3D canonical organization rather than only from a stronger LVM backbone or a specific template choice.

LVM models and canonical branch. Tab. 2(a) isolates whether the generalization gain comes from stronger LVM features or from canonical alignment. With DINOv2-S features, enabling the canonical branch raises CCPG-to-CCGR* Rank-1 from 15.5% to 17.9%, and with SAM 3D Body features it further improves transfer from 18.7% to 23.2%. This shows that reorganizing LVM features into a canonical body layout consistently improves cross-domain generalization across different appearance backbones.

3D body geometry. Tab. 2(b) replaces the default SAM 3D Body geometry with ROMP while keeping the recognition pipeline fixed. The ROMP variant remains strong with 95.2% CCPG average Rank-1, indicating that PuppetGait is not tied to a single mesh estimator. SAM 3D Body improves the average to 96.5%, suggesting that more accurate and stable human body geometry recovery benefits Puppet Alignment and GAOT, especially when body-surface correspondence is affected by clothing changes.

Sparse optimal transport. Tab. 2(c) compares hard relocation, dense OT, and the sparse TopK4 transport used in the full model. Direct relocation already reaches 22.9% CCGR* Rank-1 because it uses 3D geometry to canonicalize source tokens, but dense OT drops to 21.9%, indicating that unrestricted matching can over-mix unrelated body regions. Sparse OT recovers the best source-domain average and improves transfer to 23.2%, showing that pruned and localized soft assignment better preserves discriminative LVM semantics.

Canonical feature layout. Tab. 2(d) evaluates whether the result depends on a particular target pose or yaw. Without any canonical target layout, the SAM 3D Body LVM baseline keeps strong CCPG performance but only reaches 18.7% CCGR* Rank-1. Adding a canonical layout raises transfer to 23.0–23.2%, while keeping the walking pose without A-pose reset and changing the target yaw give close results. Thus, the main benefit comes from establishing a consistent canonical feature layout, rather than from a specific pose or view template.

5 Conclusion

We presented PuppetGait, a 3D body-aligned LVM framework for generalizable RGB gait recognition. By anchoring image-plane LVM tokens to downstream 3D body geometry and transporting them to a canonical feature layout, PuppetGait preserves rich visual semantics while introducing explicit body-part correspondence. Experiments across four benchmarks show strong within-domain recognition and consistent cross-domain gains, validating 3D-guided canonical alignment as an effective complement to stronger LVM backbones.

Limitations. PuppetGait depends on LVM based 3D body recovery, so it introduces additional computation cost. Also, extreme occlusion or severe truncation may affect the quality of correspondence. More efficient geometry predictors deserves future exploration.

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